

Jianing Tang

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PhD Student

Department of Biomedical Engineering and Radiology
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Research interests

MRI Imaging, Deep Learning, Small Vessel Disease, Phase-contrast MRI, Arterial Spin Labeling

Education

Northwestern University Chicago, IL

PhD in Biomedical Engineering 09 2022 – Present
Mentor: Dr. Lirong Yan GPA: 3.87

Northwestern University Chicago, IL
MS in Biomedical Engineering 09 2022 – 09 2025
Mentor: Dr. Lirong Yan GPA: 3.87

University of California, Irvine Irvine, CA
BS in Biomedical Engineering, minor Comp Sci 09 2017 – 06 2021
Mentors: Dr. Zhongping Chen; Dr. Andrew Browne GPA: 3.82

Honors and Awards

Prince-Meaney Translational Science Award 2026

ISMRM Neurofluid Workshop Award 3rd Place 2026

ISMRM Summa Cum Laude Merit Award 2026

ISMRM Summa Cum Laude Merit Award 2026

ISMRM Educational Stipend 2025

ISMRM Magna Cum Laude Merit Award 2025

SMRA Travel Grant 2024

MRM Editor's Pick 2024

ISMRM Educational Stipend 2023

Undergraduate Research Fellowship 2020, 2020, 2021

Publications

Zhao T,

Tang J, Moum SJ, He Y, Huang Z, Khan A, Shaibani A, Yan L. Concurrent Four-Dimensional Dynamic MRA and Perfusion Imaging Using Dual-Module Arterial Spin Labeling MRI With Stack-Of-Stars Golden-Angle Radial Acquisition. *NMR Biomed.* 2026 Sep;39(9):e70359. doi: 10.1002/nbm.70359. PMID: 42487211; PMCID: PMC13392063.

Tang J, Chung SH, Gamez M, Zhao T, Wolf MS, Pandey DK, Gorelick PB, Yan L. Pulsatility Assessment of Cerebral Perforating Arteries Using Submillimeter-Resolution Dual-VENC Phase-Contrast MRI at 3T. *J Magn Reson Imaging.* 2025 Dec 27. doi: 10.1002/jmri.70218. Epub ahead of print. PMID: 41454626.

Tang J, He Y, Zhao T, Li Z, Jin N, Bi X, Gamez MT, Yan L*. Simultaneous assessment of arterial pulsatility at different segments of cerebral arteries using multiband dual-VENC phase-contrast MRI. *NMR Biomed.* 2025 Sep;38(9):e70099. doi: 10.1002/nbm.70099.

Zhao T, **Tang J**, He Y, Gamez MT, Yan L*. Assessing cerebral microvascular pulsatility using dynamic pseudo-continuous arterial spin labeling MRI. *Magn Reson Med* 2025 Jul 22. doi: 10.1002/mrm.70004.

Zhao T, **Tang J**, Krumpelma C, Moum SJ, Russin JJ, Ansari SA, Chen Z, Feng L, Yan L*. Highly accelerated non-contrast enhanced time-resolved 4-dimensional MR angiography using stack-of-stars golden-angle radial acquisition with a self-calibrated low-rank subspace reconstruction. *Magn Reson Med.* 2025 Feb;93(2):615-629.

Tang J, Pahlavian SH, Joe E, Gamez MT, Zhao T, Ma SJ, Jin J, Cen SY, Chui HC, Yan L*, Assessment of arterial pulsatility of cerebral perforating arteries using 7T high-resolution dual-VENC phase-contrast MRI. *Magn Reson Med,* 2024 Mar 5. doi: 10.1002/mrm.30073.

Browne AW, Deyneka E, Ceccarelli F, To JK, Chen S, **Tang J**, Vu AN, Baldi PF. Deep learning to enable color vision in the dark. *PLoS One.* 2022 Apr 6;17(4):e0265185. doi: 10.1371/journal.pone.0265185. PMID: 35385502; PMCID: PMC8985995.

Li Y, Murthy RS, Zhu Y, Zhang F, **Tang J**, Mehrabi JN, Kelly KM, Chen Z. 1.7-Micron Optical Coherence Tomography Angiography for Characterization of Skin Lesions-A Feasibility Study. *IEEE Trans Med Imaging.* 2021 Sep;40(9):2507-2512. doi: 10.1109/TMI.2021.3081066. Epub 2021 Aug 31. PMID: 33999817; PMCID: PMC8834583.

Teaching experience

Teaching assistant, Department of Biomedical Engineering Spring 2021

BME 137: Introduction to Biomedical Imaging

Held discussion sessions and office hours; Generated course materials and exam questions; Graded homeworks, midterms and finals

Teaching assistant, Department of Biomedical Engineering Spring 2025

BME 306: Quantitative Systems Physiology

Held discussion sessions and office hours; Generated course materials, clicker and exam questions; Graded tests, homework and coding problems

Oral Talks

Quantitative 3D Mapping of Cardiac-Driven CSF Dynamics Using Velocity-Encoded T2-Prepared GRASE May 2026

Proceedings of the 35th Annual Meeting of ISMRM, Cape Town, South Africa

Simultaneous assessment of cardiac-driven cerebral arterial, venous, CSF dynamics and coupling with multiband dualVENC PC-MRI May 2026

Proceedings of the 35th Annual Meeting of ISMRM, Cape Town, South Africa

Cardiac-Driven Quantitative Whole-Brain CSF Dynamic Imaging Using Velocity-Encoded T2-Prepared GRASE Mar 2026

Proceedings of Fluids, Flow and Clearance in Brain Workshop of ISMRM, New Haven, USA, 2026

Simultaneous assessment of cerebral arterial, venous, and CSF flow dynamics using multiband dual-VENC phase-contrast MRI May 2025

Proceedings of the 34th Annual Meeting of ISMRM, Honolulu, USA 34, 0457

Associations of lenticulostriate artery geometry and hemodynamics with age, vascular risks, and cognitive decline: A high-resolution 7T study May 2025

Proceedings of the 34th Annual Meeting of ISMRM, Honolulu, USA 34, 1217

Quantifying Cerebral Small Vessel Geometry with Self-Supervised Shape-Aware 3D U-Net on 7T TOF MRI: Associations with Age and Cognitive Function May 2025

Proceedings of the 34th Annual Meeting of ISMRM, Honolulu, USA 34, 1294

Characterization of Cerebral Perforating Arteries using submillimeter-resolution dual-VENC PC-MRI at 3T May 2024

Proceedings of the 36th Annual Meeting of SMRA, Santiago, Chile, 2024

Simultaneous assessment of arterial pulsatility at different segments of cerebral arteries using multiband dual-VENC Phase-Contrast MRI

May 2024

Proceedings of the 33rd Annual Meeting of ISMRM, Singapore, Singapore 33, 1336

Evaluation of the robustness of carotid pulse wave velocity measurement using a single-slice oblique-sagittal phase-contrast MRI

May

2022

Proceedings of the 31st Annual Meeting of ISMRM, London, UK 31, 4962

Assessment of arterial pulsatility of cerebral perforating arteries using 7T high-resolution dual-VENC phase-contrast MRI

May 2022

Proceedings of the 31st Annual Meeting of ISMRM, London, UK 31, 3959

Assessment of arterial pulsatility of cerebral perforating arteries using 7T high-resolution dual-VENC phase-contrast MRI

March 2022

Proceedings of Ultra High Field Workshop of ISMRM, Lisbon, Portugal, 2022.

Selected Conference
Proceedings

Tang, J., Zhao, T., Gamez, M., Gorelick, P., Chui, H., and Yan, L. (2025), Assessing pulsatility of cerebral perforating arteries with 3T high-res dual-VENC PC-MRI: Development, validation, and aging evaluation. Proceedings of the 34th Annual Meeting of ISMRM, Honolulu, USA 34, 4358

Tang, J., Zhao, T., Joe, E.B., Chui, H.C. and Yan, L. (2024), Association of pulsatility of lenticulostriate arteries with cognitive decline in elderly adults: a 7T dual-VENC PC-MRI study. Alzheimer's Dement., 20: e093252. <https://doi.org/10.1002/alz.093252>

Tang, J., Zhao, T., Joe, E.B., Pahlavian, S.H., Chui, H.C. and Yan, L. (2024), Carotid stiffening with impaired damping is associated with cognitive impairment and reduced cerebral perfusion in elderly adults. Alzheimer's Dement., 20: e093187. <https://doi.org/10.1002/alz.093187>

Tang, J., Pahlavian, S.H., Joe, E.B., Turner, J., Braskie, M.N., Marmarelis, V., Chui, H.C. and Yan, L. (2023), Elevated Carotid Pulse Wave Velocity is associated with Cognitive Impairment and Cerebral Amyloid Deposition: A Phase-Contrast MRI Study. Alzheimer's Dement., 19: e079242. <https://doi.org/10.1002/alz.079242>

Tang, J., Pahlavian, S.H., Joe, E.B., Chui, H.C. and Yan, L. (2023), Carotid pulse wave velocity measured using a fast single-slice oblique-sagittal phase-contrast MRI is associated with cognitive impairment. *Alzheimer's Dement.*, 19: e066468. <https://doi.org/10.1002/alz.066468>

Skills

Python, C/C++, MATLAB, R

Solidworks, LabView, ImageJ, Radiant

Languages: Mandarin (Native), English (Professional), Japanese (Elementary)

Professional

memberships

American Heart Association 2025 – Present

Radiological Society of North America 2023 – Present

Society of MR Angiography 2023 – Present

Alzheimer's Association International Society to Advance Alzheimer's Research and Treatment 2022 – Present

International Society for Magnetic Resonance in Medicine 2022 – Present